

Chengyuan Xu

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AREAS OF INTEREST

Vision-Language Models & Multimodal Generative Models – video understanding, data processing, ablation, training, finetuning, and inference-time optimization

Segmentation, Multi-object Tracking, Human-AI Collaboration, Interactive Machine Learning, Augmented Reality

EDUCATION

2017–24 Ph.D. in Media Arts and Technology, University of California, Santa Barbara
Committee: [Tobias Höllerer](#), [Jennifer Jacobs](#), [Marko Peljhan](#), [Curtis McCully](#)

2020–24 M.S. in Computer Science, University of California, Santa Barbara
Advisor: [Tobias Höllerer](#)

PROFESSIONAL EXPERIENCE

2023– **Adobe Firefly**
Senior Applied Scientist

2026 VLM and LLM

- In-house VLM and LLM models to improve training and inference data.
- Building agentic workflows for self-improving prompt rewriting optimization.

2025 VLM-based video data pipeline

- Co-designed Firefly's in-house video data understanding and moderation pipeline from 0 to 1, reducing training preparation time for 100 millions of videos from months to days.

2025 [Adobe Firefly multimodal foundation model](#)

- Core contributor. Data processing, ablation, mid-training, and post-training experiments for Adobe's multimodal diffusion foundation model.

2024 [Adobe Firefly video model training](#)

- Core member of Firefly's first video diffusion model training for public beta. Designed and trained the flickering-free video decoder to unblock product release.

Research Scientist/Engineer Intern, Summer Internship

2023 [Internship Model Shipped in Adobe Product](#)

- GAN-based facial editing model. Internship model shipped worldwide in Photoshop Express.

- 2022 **Appen Limited**
Computer Vision Intern, Dec. 2021 – Sept. 2022
- We proposed a AI-in-the-loop video annotation workflow that helped human annotators track faces in very long videos 30% faster with better quality.
 - We designed a large user study and investigated 3,466 person-hours of annotation work in “[Comparing Zealous and Restrained AI Recommendations in High-stakes Human-AI Collaboration](#).” The analysis revealed significant findings to guide future designs of human-AI collaboration systems in high-stakes tasks.
- 2021 **Benioff Ocean Science Laboratory**
Computer Vision Researcher, Summer Internship
- A trash detection model and a dataset “[BOI Baltimore Trash Wheel Computer Vision Model and Dataset](#)”, deployed in Baltimore to identify 15 types of river waste. The work collects data and assist marine scientists in better understanding the types and sources of ocean-bound river waste.
- 2019 **Las Cumbres Observatory**
Imaging Intern, Summer Internship
- First author of top astrophysical journal paper “[Cosmic-ConNN](#).” The segmentation model was deployed to 20+ astronomical telescopes globally, processing thousands of space observations daily.

PUBLICATIONS

- 2025 Zhu, Z., Duarte, K., · Rizve, MN., **Xu, C.**, Kalarot, R., & Yuan, J. “CompSlider: Compositional Slider for Disentangled Multiple-Attribute Image Generation.” [IEEE ICCV 2025 proceedings](#).
- 2024 **Xu, C.**, Kumaran, R., Stier, N., Yu, K., & Höllerer, T. “Multimodal 3D Fusion and In-Situ Learning for Spatially Aware AI.” [IEEE ISMAR 2024 oral presentation](#).
- 2023 Zhu, J., Kumaran, R., **Xu, C.**, & Höllerer, T. “Free-form Conversation with Human and Symbolic Avatars in Mixed Reality.” [IEEE ISMAR 2023 oral presentation](#).
- 2023 **Xu, C.**, Lien, K.C., & Höllerer, T. “Comparing Zealous and Restrained AI Recommendations in High-stakes Human-AI Collaboration.” [ACM CHI 2023 oral presentation](#).
- 2022 **Xu, C.**, Dong, B., Stier, N., McCully, C., Howell, D. A., Sen, P., & Höllerer, T. “Interactive Segmentation and Visualization for Tiny Objects in Multi-megapixel Images.” [CVPR 2022 demo and proceedings](#).
- 2022 **Xu, C.**, McCully, C., Dong, B., Howell, D. A., & Sen, P. “Cosmic-CoNN: A Cosmic Ray Detection Deep Learning Framework, Dataset, and Toolbox.” [240th American Astronomical Society meeting, oral presentation](#).
[The Astrophysical Journal](#), Volume 942, Number 2.
- 2021 Hiramatsu, D. et al., including **Xu, C.** “The electron capture origin of supernova 2018zd.” [Nature Astronomy, cover story Volume 5, Issue 9](#).

SERVICE

- 2025 Reviewer, IEEE ISMAR 2025, Experimental Astronomy Journal
- 2024 Reviewer, ACM CHI 2024, ACM IMX 2024, ACM VRST 2024, IEEE ISMAR 2024, IEEE TCSVT
- 2023 Reviewer, ACM CHI PLAY 2023, IEEE PacificVis 2024
- 2022 Reviewer, ACM CHI 2023
- 2021–22 Student Representative, Media Arts and Technology Program, UCSB.
- 2020–21 Peer Mentor, Women In Computer Science, WiCS Mentorship Program, UCSB.

ACADEMIC EXPERIENCE

- 2023-25 SIGCHI member, the ACM Special Interest Group on Computer-Human Interaction
- 2021 SDSC Cyberinfrastructure-Enabled Machine Learning Summer Institute
- 2018–23 Graduate Student Researcher, University of California, Santa Barbara

GRANTS AND AWARDS

- 2018-22 International Doctoral Recruitment Fellowship (\$15,000 Annually).
- 2020 Mellichamp 21st Century Global Dynamics Graduate Research Fellowship (\$7,500).
- 2018 Media Arts and Technology Grant (\$2,500).
- 2018 MAT End of Year Show Grant (\$750).